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## Self-regulating small wind turbine generator system and method


### Abstract

A small turbine generator is connected to a wind turbine. The wind turbine spins freely due to a variable wind force on the wind turbine. The generator includes a spinning rotor that is responsive and spins proportional to spin of the wind turbine. A stator is connected to an actuator. The stator is selectively moved by the actuator nearer to or further from the rotor to vary the electrical power generated. Movement of the stator is controlled such that an electrical measure, such as voltage of the electrical power generated, does not exceed a threshold level.



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## Background/Summary

TECHNICAL FIELD

(1) The invention generally relates to wind energy and wind turbines, and more particularly relates to a self-regulating generator for a small wind turbine that is able to adjust output notwithstanding excess wind speed.

BACKGROUND

(2) Wind energy is prolific. Energy from wind is thought to be cleaner and more abundant than traditional fossil fuels. Energy is gathered from wind typically by a bladed propeller of a wind turbine. As the propeller of the wind turbine rotates with the wind, an electricity generator is caused

to rotate creating electricity.
(3) An alternating current (A/C) electricity generator typically includes a stator, which is a stationary magnetic field. Within the magnetic field of the stator, an armature on a rotating shaft of the generator is caused to rotate by the turning of the wind turbine. The armature is, for example, a coiled copper or other metal wire. The shaft rotates on slip rings connecting respective ends of the wire of the armature. Respective conductive brushes of the generator contact the slip rings, respectively, to carry generated electricity on to a grid or circuit.
(4) Because of wind speed and direction variation due to weather, voltage output from a wind turbine generator must be regulated. Regulation of the voltage of an electricity generator powered by wind has conventionally been effected by turning off or restricting the wind propeller rotation, varying blade angle or displacement of the propeller, or sending the generated electricity through resistor bank(s). These means for regulation of voltage have been problematic for a number of reasons.
(5) Turning off or restricting the wind propeller rotation and varying blade angle or display each require mechanical complexity which is subject to malfunction, destruction, noise, vibration, or other issues. For example, extreme high winds can cause the propeller and other mechanical components damage. Further, if too high a voltage or amperage is output from the generator, downstream controls can be damaged and/or resistor banks or other components are required to discharge load from the turbine generator.
(6) Conventionally, small wind turbines, which may be employed in residential homes, remote areas or otherwise, have controlled the generator by slowing the wind turbine rotation. This has often been accomplished by mechanically affecting the wind turbine. For example, tail yawing may be employed, in which the tail of the turbine turns the turbine from direct wind. However, this can create sudden noises and vibration because blades are suddenly cutting the wind at different angle. In other instances, blade furling may be employed in which blades are designed to self-distort in high winds. This varies aerodynamics of the wind turbine blades to change rotation speed and can consequently create noise and vibration. Yet other mechanical adjustment may be made to the blade pitch. Mechanical and/or electrical control mechanisms make the adjustments. In any event, control systems of small wind turbine generators that are suitable to avoid or limit problems have been complex, expensive and subject to damage or the like.
(7) Small wind turbines are particularly susceptible to problems. The attendant cost and complexity of small wind turbines and generators has limited wider adoption and acceptance of the systems. Moreover, these systems have typically been higher priced than competing solar and other electrical energy generating sources. Wider adoption of small wind turbines and electricity generators is desirable for the environment. Also, these systems meet needs of those without access to other options or who favor clean energy source.
(8) It would therefore be a significant benefit and improvement in the art and technology to provide wind turbines and generators, for example, small wind turbine generators, that overcome these and other problems and limitations. It would also be a significant benefit and improvement to provide lower cost and less complexity of these systems. It would be even further significant benefit and improvement to provide small wind turbine generators which are desirable in terms of costs, complexity, mechanics, noise, and other aspects.

SUMMARY

(9) An embodiment of the invention includes a system. A wind energy drives a propeller. The system includes a rotor that spins responsive to the propeller and a stator that is movable with respect to the rotor. The stator is moved to obtain a desired generated electrical output.
(10) Another embodiment of the invention is a small turbine generator. The generator is connected to a wind turbine. The generator outputs a generated electricity. The wind turbine spins freely. The small turbine generator includes a spinning rotor responsive and spinning proportional to spin of the wind turbine, a centrifugally static stator, an actuator connected to the stator, a guide shaft

connected to the actuator, and a slider connected to and electrically shielded from the stator and slidably connected to the guide shaft. The actuator moves the stator on the guide shaft, in respect of the spinning rotor, to vary a voltage of the generated electricity.
(11) Yet another embodiment of the invention is a method. The method includes moving a stator toward a magnetic field of a rotor if a voltage level of electricity generated is below an excess voltage level and moving the stator away from the magnetic field of the rotor if the voltage level of electricity generated approaches the excess voltage level.
(12) Another embodiment of the invention is a method of operation of a small wind turbine generator. The small wind turbine generator is connected to a small wind turbine. The method includes rotating a rotor of the generator responsive and in steady proportion to spin of the small wind turbine, providing a stator in vicinity of a magnetic field of the rotor, generating an electric current of a voltage, signaling an actuator connected to the stator, the actuator positions the stator within the magnetic field of the rotor, and controlling the signaling to cause the actuator to position the stator closer to the magnetic field of the rotor to increase the voltage of the electric current and farther from the magnetic field of the rotor to limit the voltage of the electric current on reaching a threshold voltage.
(13) Yet other embodiments of the invention include methods of manufacture of the embodiments of systems.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present invention is illustrated by way of example and not limitation in the accompanying figures, in which like references indicate similar elements, and in which:
(2) FIG. **1** illustrates a left side and partial cutaway view of a conventional wind turbine and wind turbine generator;
(3) FIG. **2** illustrates a left and top side perspective and partial cutaway view of a conventional wind turbine generator;
(4) FIG. **3** illustrates a left side exploded view of conventional gears and wind turbine generator of FIG. **1**;
(5) FIG. **4** illustrates a wind turbine generator connected to a wind turbine, according to certain embodiments of the invention;
(6) FIG. **5** illustrates an example of the wind turbine generator connected to the wind turbine of FIG. **4**, according to certain embodiments of the invention;
(7) FIG. **6** illustrates a method for operating a wind turbine generator connected to a wind turbine, according to certain embodiments of the invention;
(8) FIG. **7** illustrates a method of varying position of a stator to a rotor in a wind turbine generator for varying output electrical power of the generator, according to certain embodiments of the invention;
(9) FIG. **8** illustrates a wind turbine generator system including a stator positionable in relation the a rotor for desired electrical output power, according to certain embodiments of the invention;
(10) FIG. **9** illustrates test results obtained with an example wind turbine generator system, according to certain embodiments of the invention;
(11) FIG. **10** illustrates other test data obtained with an example wind turbine generator system, according to certain embodiments of the invention; and
(12) FIG. **11** illustrates certain power output of an example wind turbine generator at various wind speeds, according to certain embodiments of the invention.
DETAILED DESCRIPTION
(13) Referring to FIG. **1**, a system **100** includes a wind turbine **102**. The wind turbine **102** includes

a propeller **104** connected to one or more gear **106**. The gear **106** is connected to an electric generator **108**. The wind turbine **102** may be rotatingly or otherwise mounted, for nonexclusive example, on a tower **110** or other mount. The wind turbine **102** may, but need not necessarily, include a vane (not shown) or other mechanism for rotating or moving the propeller **104** into a desired position for wind collection. For nonexclusive example, the propeller **104** is positioned to collect wind A. To vary voltage output by the electric generator **108**, the propeller **104** is mechanically adjusted or otherwise controlled and/or excess voltage is off loaded by electric circuits and elements (not shown).

(14) Referring to FIG. **2**, in conjunction with FIG. **1**, an electric generator **200**, such as, for nonexclusive example, the electric generator **108**, includes a stator **202** and a rotor or armature **204**. The stator **202** presents a stationary magnetic field. As nonexclusive example, the stator **202** is one or more magnet with copper windings. A rotating shaft **206** connects the armature **204** to one or more gear, such as for nonexclusive example, the gear **106** or other mechanical translator, and a rotating assembly, such as for nonexclusive example, a propeller **104** or other rotating device driven by the wind A.

(15) The armature **204** includes, for nonexclusive example, field windings **208** of coil that, when moving within the magnetic field of the stator **202**, generate electricity. One or more fixed support or frame **210** supports the armature **204**. Bearings or slip rings **212** connected to the support **210** connect to the armature **204** to reduce friction of the rotating armature **204**. A commutator **214** is connected to the armature **204** for conducting and collecting electricity. A brush assembly (not shown) of one or more brush contacts the commutator **214** to send generated electrical current on to a load, such as for nonexclusive example, a home electrical circuit and consequent elements for stabilizing and off-loading voltage. The voltage output from the generator **200** has been regulated in the manner previously described, primarily by varying rotation speed of the propeller **104** or other rotating device driven by the wind A.

(16) Referring to FIG. **3**, in conjunction with FIGS. **1** and **2**, a gear assembly **302** connects to the generator **200** and to a rotating shaft **304** connected to a rotating device, such as for nonexclusive example the propeller **104**. The gear assembly **302** may include one or more gear to translate the mechanical rotation of the propeller **104** or other rotating device driven by the wind A. The extent of translation of the mechanical rotation may be varied by virtue of the gear assembly **302** according to the application. In general, the gear assembly **302** for a small wind turbine **102** must translate mechanical rotation of the propeller **104** to rotate the rotating shaft **206** of the generator **200** in manner for generation of suitable electric output voltage of the generator **200**. As nonexclusive example, an output voltage of about 80 V.sub.dc or less may be desired for a small wind turbine **102** servicing a home or residence.

(17) Problems of excess output voltage can be presented in the foregoing because of several factors, as was previously discussed. These problems stem largely from the mechanical and electrical facilities that are presented. Mechanical features for varying the propeller speed, such as tail yawing, blade furling and blade pitch control, require mechanical movements or changes that are subject to damage or destruction and other obsolescence or issue. Moreover, there are limitations to the effectiveness of these mechanical features to vary propeller speed, particularly in high wind speeds. This results in requirements of electrical features to manage and/or off-load output power from the generator. Control systems for these environments can be complex and expensive. In any event, output power must be regulated in some manner to prevent overloads.

(18) Referring to FIG. **4**, a system **400** includes a wind turbine **402**. The wind turbine **402** includes a propeller **404** connected to a rotating shaft **406**. The rotating shaft **406** is connected, for nonexclusive example, through one or more connected gear (not shown) as applicable, to a generator **408**. The rotating shaft **406** is supported by a support **407**.

(19) The generator **408** includes a stator **410** and an armature **412** connected to an actuator **414**. The generator **408** includes a frame **414** for retaining the stator **410** and the armature **412** in desired

relationship. In the generator **408**, the actuator **414** varies position of the armature **412** with respect to the stator **410** to consequently vary the output power of the generator **408**. The stator **410** may, as nonexclusive example, include one or more magnet and coil winding. The actuator **414** may, as nonexclusive example, include a wound coil of the armature **412** that moves via the actuator **414** with respect to the stator **410**, and rotates with wind against the propeller **404** outputting electricity. In effect, the actuator **414** changes the relationship of the armature **412** to the magnetic field of the stator **410**. The opposite is also possible, such that the armature **412** includes one or more magnet and coil winding creating the magnetic field and the stator **410** includes the wound coil outputting electricity.

(20) In any event, the armature **412** is variably positioned by the actuator **414** in relationship to the stator **410** in order to desirably vary the output voltage of the generator **408**. This arrangement allows the propeller **404** to rotate without necessity of braking as via the problematic conventional mechanisms and operations. Moreover, this arrangement provides a suitable control of output electrical power from the generator **408** by virtue of the variable position of the armature **412** in the magnetic field of the stator **410** (or vice versa, as applicable in the embodiment).

(21) Referring to FIG. **5**, a system **500** includes a wind turbine **502** connected to an electrical generator **504**. The wind turbine **502** includes a bladed propeller **506** connected to a turbine shaft **508**. The bladed propeller **506** rotates due to wind forces and rotates the turbine shaft **508**. Although not shown in FIG. **5**, the turbine shaft **508** may connect to one or more gear to translate the rotational motion of the turbine shaft **508**.

(22) The turbine shaft **508**, or if one or more gear then the gear(s), is connected to a generator shaft **510**. The generator shaft **510** rotates in cooperation with the turbine shaft **508**, either directly or through the one or more gear. A support **512** connects to a shaft bearing **514** through which the generator shaft **510** connects and rotates. The generator shaft **510** connects to a rotor **516** that spins with the generator shaft **510**. The generator shaft **510** may additionally connect to another support **518** and another bearing **520** of the other support **518**. In this manner, the rotor **516** spins in accordance with rotation of the generator shaft **510**, whatever the wind force may be to the bladed propeller **506** correspondingly spin is imparted thereby to the generator shaft **510** and thus the rotor **516**.

(23) The rotor **516** is, includes or is connected to one or more magnet **517**. Spin of the rotor **516** consequently spins the magnet **517**. A stator **522** is selectively positionable within a magnetic field of the rotor **516**. The stator **522** is, includes or is connected to one or more coil winding **523**. The stator **522** may be selectively moveable relative to the magnetic field of the rotor **516**. In certain non-exclusive embodiments, the stator **522** connects to one or more, for non-exclusive example two, four or other number, slide shaft **524**. The slide shaft **524** is connected between the supports **512**, **518**. In the embodiment, the stator **522** is slidably connected to the slide shaft **524**, for non-exclusive example, by one or more slide bearing, bushing or similar slider **526**. The slider **526** is electrically shielded from the coil winding **523** of the stator **522**.

(24) An actuator **530** is connected to a bracket **528**. The bracket **528** is connected to the stator **522**, for non-exclusive example, through pathways of the support **518**. The actuator **530** coupled to the bracket **528** pushes the stator **522** along the slide shaft **524**. The stator **522**, connected to the bracket **528**, is pushed into and out of the magnetic field of the rotor **516** as it spins. Electrical current is generated in the coil winding **523** of the stator **522** when the stator **522** is positioned in the magnetic field of the rotor **516**. This electrical current is output by the coil winding **523** to one or more wire **532**, for non-exclusive example, three wires if 3-phase electrical current. A voltage output on the wire **532** is variable by displacing the extent of the coil winding **523** within the magnetic field of the rotor **516**.

(25) In certain nonexclusive embodiments, the electrical current of the coil winding **523** electrically connects by supply wires **534** to a bridge rectifier **536**. The rectifier **536** is for nonexclusive example a diode that converts AC to DC power or similar elements. Voltage of the DC power from

the rectifier **536** is electrically connected by output wires **540** to a voltage reducer **538**. The reduced voltage signal from the voltage reducer **538** is electrically connected to the actuator **530** by actuator power wires **542**. The actuator **530** is controlled to stroke the stator **522** position by another reduced voltage signal on wires **546**. Another voltage reducer **544** electrically connected to the wires **546** is also electrically connected by the output wires **540** to the rectifier **536**.

(26) In operation, the rotor **516** spins according to rotation of the propeller **506** driven by wind energy. The stator **522** is selectively positioned within the magnetic field of the rotor **516** in order to obtain a desired output generated voltage. For nonexclusive example, greater voltage may be obtained when the stator **522** is positioned more in the magnetic field of the rotor **516** and lesser voltage may be obtained when the stator **522** is positioned less in the magnetic field of the rotor **516**. The stator **522** is shifted in position by the actuator **530** and, if applicable, any bracket **528**. The stator **522** is maintained from spinning by the slide shaft **524** and slider **526**, or other frame or structure for selective displacement. As such electricity is generated by relation of the stator **522** to the rotor **516** spinning by wind force on the propeller **506**, the generated electricity is output on the wires **532** and **534** or otherwise as desired in the application.

(27) In certain nonexclusive embodiments, the systems and methods operate to generate electricity of desired voltage. For example, the propeller **506** may spin at whatever rate and the voltage of output electricity may be regulated by selective and moveable positioning of the stator **522** in relation to the magnetic field of the rotor **516**. If the propeller **506** spins at a fast rate because of excessive or high wind, which could provide excessive voltage of output electricity, the stator **522** may be moved within the magnetic field of the rotor **516** to adjust the voltage output. This allows stabilization of voltage at about maximum (or other measure, as desired) without varying, braking, or otherwise distorting the propeller **506** or its rotation and without excessive or complex voltage offloading elements. Control system(s) for the positioning of the stator **522** in relation to the magnetic field of the rotor **516** can be widely varied as applicable for the embodiment and application.

(28) Referring to FIG. **6**, a method **600** includes providing a rotor **602**. The rotor spins responsive to an energy source, for nonexclusive example the wind. The rotor includes a first part of an electricity generator. The method **600** further includes providing a stator **604** for electrical generation interaction with the rotor. The stator includes a second part of an electricity generator. The method **600** also includes variably positioning the stator **606** in select spatial relationship with the rotor. The method **600** further includes locating the stator **608** by the variably positioning the stator **606** to obtain selective generation of electricity. By locating the stator **608** in position with respect to the rotor, the electricity generated is of a desired voltage, for nonexclusive example, voltage is limited to a maximum.

(29) Referring to FIG. **7**, a method **700** for converting wind energy to an electrical current includes providing a wind propeller **702** and providing a magnetic rotor **704** connected to the wind propeller. The method **700** also includes receiving the wind energy to the wind propeller **706** causing the wind propeller to spin. The method further includes spinning the magnetic rotor **708** responsive to wind force turning the wind propeller.

(30) The method **700** additionally includes providing a moveable stator **710** for interacting within a magnetic field of the rotor. The moveable stator includes coil windings. The moveable stator is selectively positioned for obtaining select electrical current, such as for nonexclusive example a desired voltage or desired voltage limit. The method **700** may, but need not necessarily, also include positioning the moveable stator **712** to limit voltage of the electrical current to a maximum voltage measure.

(31) The method **700** additionally includes controlling the position of the moveable stator **714** to allow the wind propeller to freely rotate without variation yet maintain the desired voltage limit of the electrical current generated. The method **700** may, but need not necessarily, include providing an actuator **716** connected to the moveable stator and signaling the actuator **718** to reposition the

moveable stator based on the wind energy, for nonexclusive example, based on the speed of rotation of the wind propeller due to the wind energy. The method may, but need not necessarily, also include providing a fixed shaft **720** slidably connected to the moveable stator. The moveable stator slides along the fixed shaft responsive to the signaling the actuator **718**, to position the moveable stator in select desired position within the magnetic field of the rotor.

(32) Controlling the position of the moveable stator **714** may, but need not necessarily, include controlling by one or more computer or microprocessor. The control may but need not necessarily include options for control that may be manually or otherwise input or directed, as well as automated. In certain nonexclusive embodiments, the wind propeller is forced to spin by wind energy at spin rate that is not controlled or is only minimally controlled. In such instance, the moveable stator is selectively positioned in the magnetic field of the spinning rotor in order to regulate the electricity output. For nonexclusive example, the particular positioning of the moveable stator is determined according to limiting the output voltage to a maximum measure.

(33) Referring to FIG. **8**, a system **800** includes an electric generator **802**. The generator **802**. The generator **802** includes a rotor **804** and a stator **806**. The rotor **804** spins as result of an energy source, such as for nonexclusive example, a shaft **810** connected, directly or indirectly, to the rotor **804** and to a spinnable wind turbine blade **808** or other element receiving wind energy or other energy source. The stator **806** is positionable with relation to the rotor **804**.

(34) The stator **806**, for nonexclusive example, is connected to a linkage **812**. The linkage **812** is communicatively connected to an actuator **814**. An output electrical signal **816** and a control electrical signal **818** are, respectively, communicatively connected to the stator **806**. The control electrical signal **818** is connected to a controller, for nonexclusive example, including a converter **820** communicatively connected to a regulator **822** and to a reducer **824**. The controller is communicatively connected to the actuator **814**.

(35) For nonexclusive example, the stator **806** includes a coil winding **826** and the rotor **804** includes a magnet **828**. Alternatively, the coil winding **826** may be included in the rotor **804** and the magnet be included in the stator **806**. In any event, and whatever the stator **806** and rotor **804** may include, rotation of the rotor **804** in relation to the stator **806** creates the output electrical signal **816** and provides the control electrical signal **818**.

(36) As the rotor **804** spins, the stator **806** is positioned by the actuator **814** and the linkage **812** to vary a parameter or characteristic of the output electrical signal **816**, such as for nonexclusive example to vary voltage output of the output electrical signal **816**.

(37) In certain nonexclusive embodiments, the stator **806** may ride on a guide **826** via bushing(s), bearing(s) or similar slider(s) **828** or other locating structures. In such nonexclusive embodiments, the guide **826** may be connected to a structure(s) **830**. Of course, a variety of elements are possible for guiding the stator **806** in relation to the rotor **804**. And in alternatives, the rotor **804** may be selectively moveable and positionable with respect to the stator **806**.

(38) Control of the actuator **814** may include the same or additional elements to those of the converter **820**, the regulator **822** and the reducer **824**. In certain nonexclusive embodiments, control of the actuator **814** is provided by a feedback mechanism or device in order to obtain a desired parameter or characteristic of the output electrical signal **816**. Varying position of the stator **806** in relation to a magnetic field of the rotor **804**, or vice versa, creates the desired output electrical signal or other output.

(39) In operation, the rotor spins because of an energy source thereto, for nonexclusive example, wind on a propeller. The stator is moved towards and away from the rotor to obtain a desired characteristic of an output energy according to rotation of the propeller **506** driven by wind energy. As the stator is moved towards the rotor, voltage or other characteristic of the output energy is increased, and as the stator is moved away from the rotor, voltage or other characteristic of the output energy is decreased. By varying the position of the stator with respect to the rotor (or vice versa), a desired output energy may be obtained. As nonexclusive example, that output energy may

have an excess threshold that is limited in the output energy through the operations.

EXAMPLE

(40) In an example nonexclusive embodiment in accordance with the foregoing, a wind propeller is connected, directly or indirectly through gears, to a first electromagnetic part, such as a magnetic rotor providing a spinning magnetic field(s). A second electromagnetic part, such as a stator of coiled wire, is selectively positioned within the magnetic field(s) in order to maintain a maximum voltage level of about 70+/−volts of output electricity notwithstanding that the wind propeller may turn from excessive wind.

(41) Test results **900** obtained for this example are shown in FIG. **9**.

(42) Where the left graph shows a conventional wind turbine generator output of electricity under wind turbine RPM rates, and the right graph shows a voltage output limit achieved by varying position of a stator with respect to a rotor under the same wind turbine RPM rates.

(43) Other test data **1000** for the example is included in FIG. **10**.

(44) Where "No Tech" indicates results of conventional wind turbine and wind turbine generator performance, and "W/Tech" indicates results of certain of the present embodiments.

(45) Also, power output at various wind speeds measured for certain of the present embodiments is profiled **1100** in FIG. **11**.

(46) As will be understood, wide variation is possible in the foregoing embodiments. Although the rotor in certain embodiments includes magnets for a magnetic field and spins, the rotor could alternately be a spinning coil wire that electrically connects, such as by brushes or otherwise, to output electrical wires or elements. In such instance, the stator may include magnets rather than coil wire, and the magnetic field may be from the stator that is moveable with respect to the rotor to obtain a desired output electricity. Although three phase output AC electricity is generated in the foregoing embodiments, any other electricity that is generated in similar manner via moveable stator or rotor, as the case may be, within magnetic field is possible.

(47) Though wind energy is described, other renewable or non-renewable energy sources, may drive the turbine generator, as applicable. For nonexclusive example, flowing water may provide the energy source. Specific elements of the wind turbine generator and its action are also subject to wide variation. Any similar self-regulating generator, of a sliding stator or rotor moveable within a magnetic field, can be employed to provide more or less electrical power, as per the applicable environment and application. For example, an actuator for moving the stator or rotor, as applicable, can be any of electric, electronic, pneumatic, hydraulic, spring, gas piston, or other. Also, for example, signal to trigger the actuator to move the stator or rotor, as applicable, can be any of electric, electronic, feedback, pneumatic, hydraulic, spring, digital, optical, laser, artificial intelligence, telepathic, or otherwise.

(48) Although not detailed, housing of the unit or units of the turbine and/or the turbine generator can be contained in single or multiple housing or modules, and parts or all of the respective elements and units may be varied in any manner. Although housing of the turbine generator unit is illustrated as standalone, the separate parts or elements can be integrated into or with other devices or systems. Software, apps, or the like are possible for communication and control or otherwise, in the embodiments. Variation is also possible in the operations of the turbine generator and its scale. Although certain operations and elements for operation are disclosed, numerous other steps, operations, processes and methods, as well as other and similar elements, may be implemented in the systems.

(49) In the foregoing, the invention has been described with reference to specific embodiments. One of ordinary skill in the art will appreciate, however, that various modifications, substitutions, deletions, and additions can be made without departing from the scope of the invention. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications substitutions, deletions, and additions are intended to be included within the scope of the invention. Any benefits, advantages, or solutions to problems

that may have been described above with regard to specific embodiments, as well as device(s), connection(s), step(s) and element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced, are not to be construed as a critical, required, or essential feature or element.

## Claims

1. A turbine generator, the generator is connected to a wind turbine with a propeller, the generator outputs a generated electricity, the wind turbine spins freely, comprising: a spinning rotor responsive and spinning proportional to spin of the wind turbine; a stator moveable along a linear axis relative to the rotor; an actuator connected to the stator and configured to position the stator relative to the rotor along the linear axis; a guide shaft connected to the actuator; a bearing connected to the stator, the bearing is electrically shielded from the stator, and slidably connected to the guide shaft; and a controller communicatively connected to the actuator and configured to continuously adjust the position of the stator along the linear axis in real-time based on measured electrical output; wherein the controller receives a voltage measurement of the generated electricity and adjusts the actuator: to move the stator towards the rotor along the linear axis if the voltage is below a threshold level; and to move the stator away from the rotor along the linear axis if the voltage exceeds the threshold level, thereby dynamically regulating output voltage without requiring mechanical braking or electrical dumping; and the stator adjusts independent of speed variations of the rotor, enables voltage regulation without braking the wind turbine or propeller dynamics level and move the stator out of relationship with the rotor if the measured exceeds the threshold voltage level.
2. The turbine generator of claim 1, wherein the stator includes a coil winding and the rotor includes a permanent magnet forming a magnetic field, wherein the actuator controls position of the stator to optimize the magnetic flux interaction for consistent electrical generation.
3. The turbine generator of claim 2, wherein the controller includes a processor and performs according to programming of the processor for signaling the actuator to move.